Two Fresh Frozen Evaluations

Independent cohorts • no URL/hostname overlap • identical models and thresholds

Cohort	Condition	Accuracy	Precision	Recall	F1	ROC-AUC	PR-AUC	FPR	FNR
5% phishing	Real-only	97.35%	73.93%	72.60%	73.26%	93.11%	78.35%	1.35%	27.40%
5% phishing	+10% GAN	97.75%	78.95%	75.00%	76.92%	92.98%	80.91%	1.05%	25.00%
10% phishing	Real-only	97.20%	86.66%	85.10%	85.87%	97.03%	91.34%	1.46%	14.90%
10% phishing	+10% GAN	97.84%	89.92%	88.30%	89.10%	97.45%	93.51%	1.10%	11.70%

Stable finding

Across both independent cohorts, GAN augmentation improved accuracy, precision, recall, F1 and PR-AUC while reducing both FPR and FN

ROC-AUC was mixed: -0.13 points on the 5% cohort and +0.41 points on the 10% cohort.

The higher 10% recall is not caused by prevalence itself: the second cohort contains a different, older active-phishing slice that pr

Augmentation Deltas

